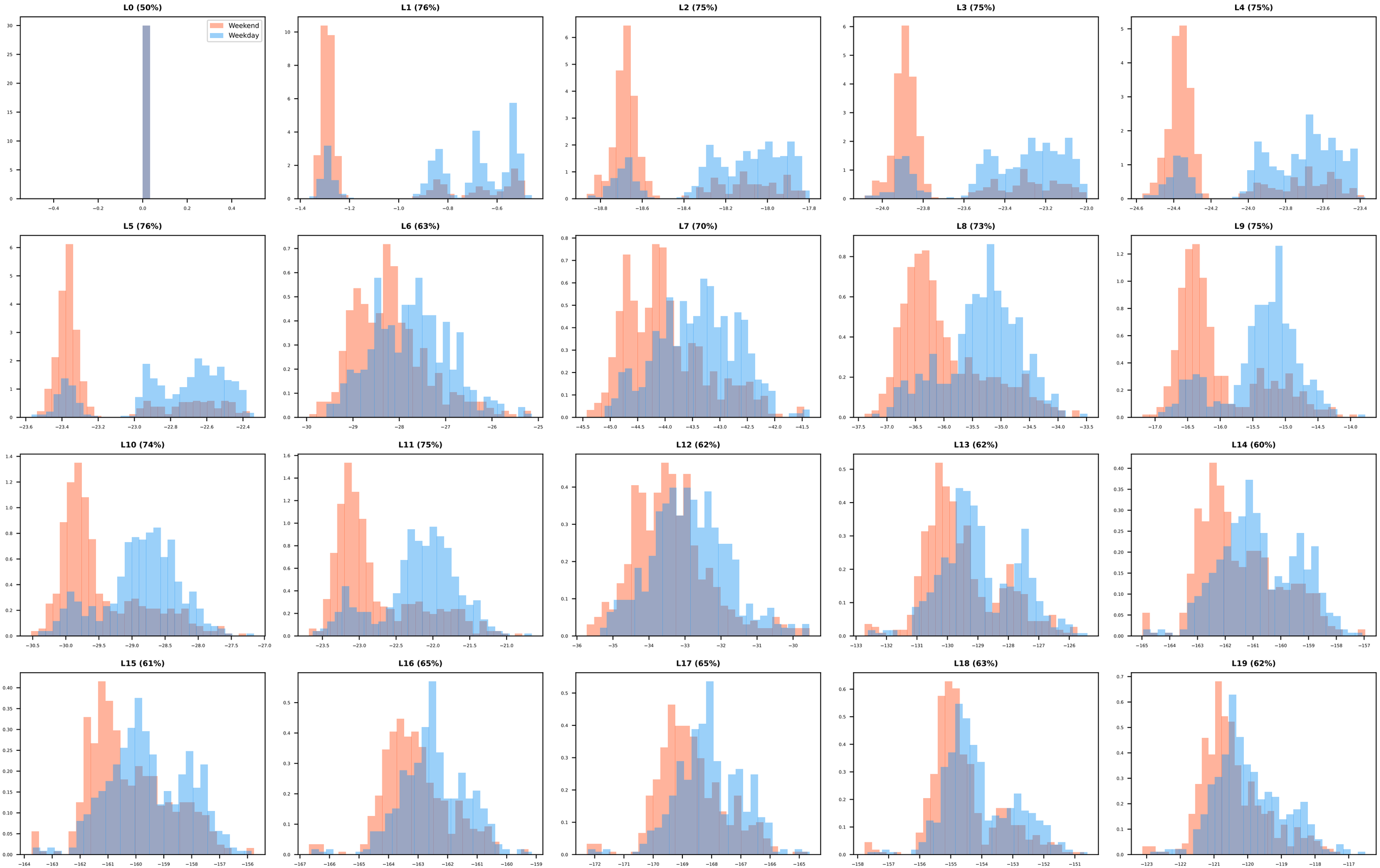
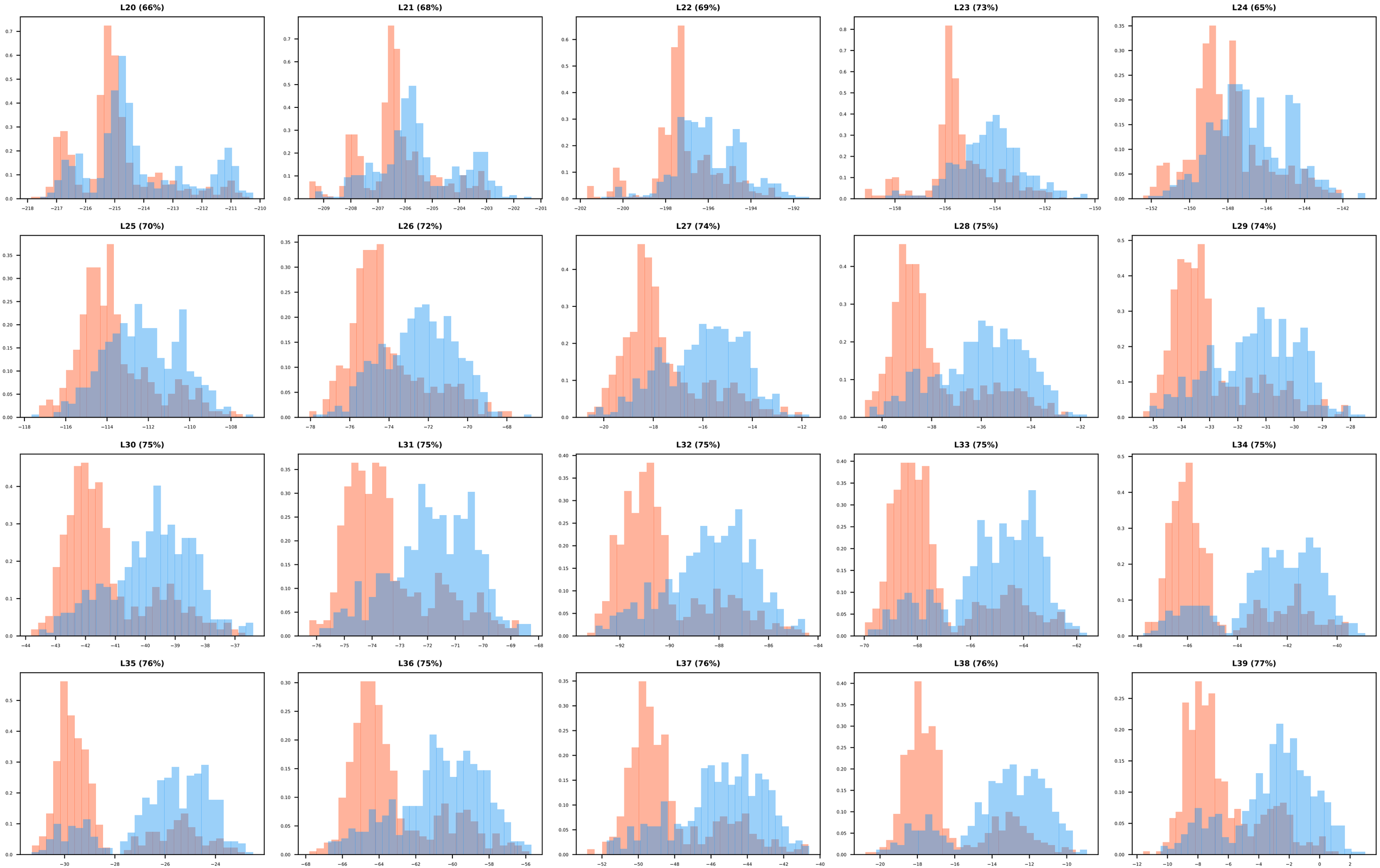
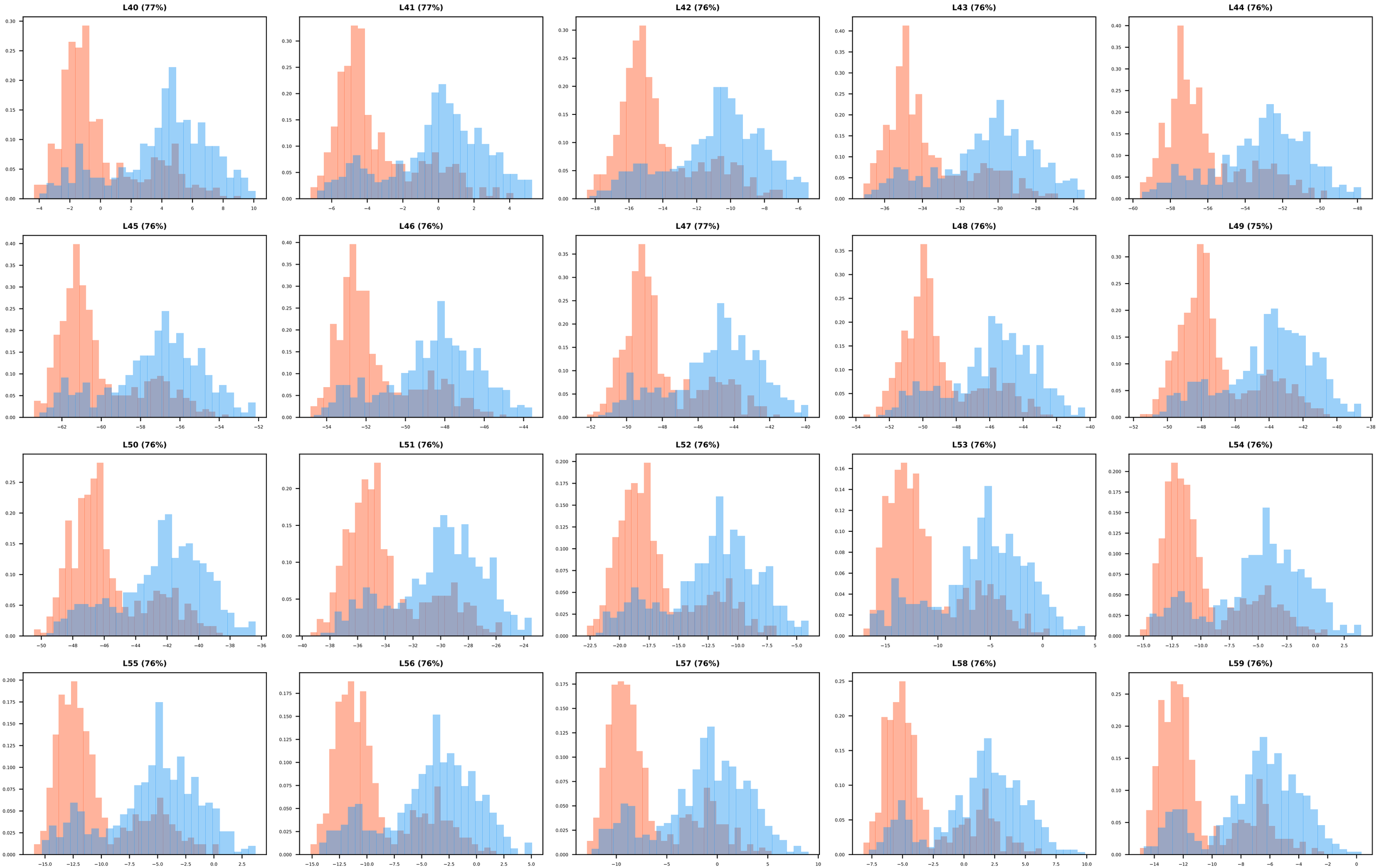
Projection Distribution — V8/exp1a/cfg2 (geo=orig_prompt=cot) — Page 1



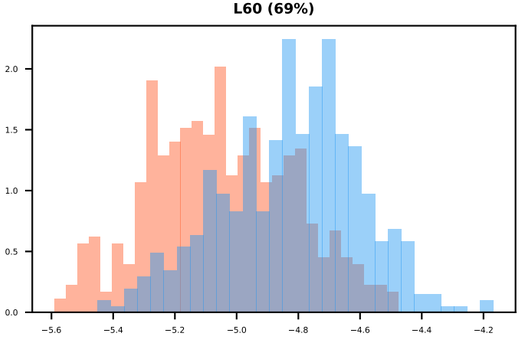
Projection Distribution — V8/exp1a/cfg2 (geo=orig_prompt=cot) — Page 2



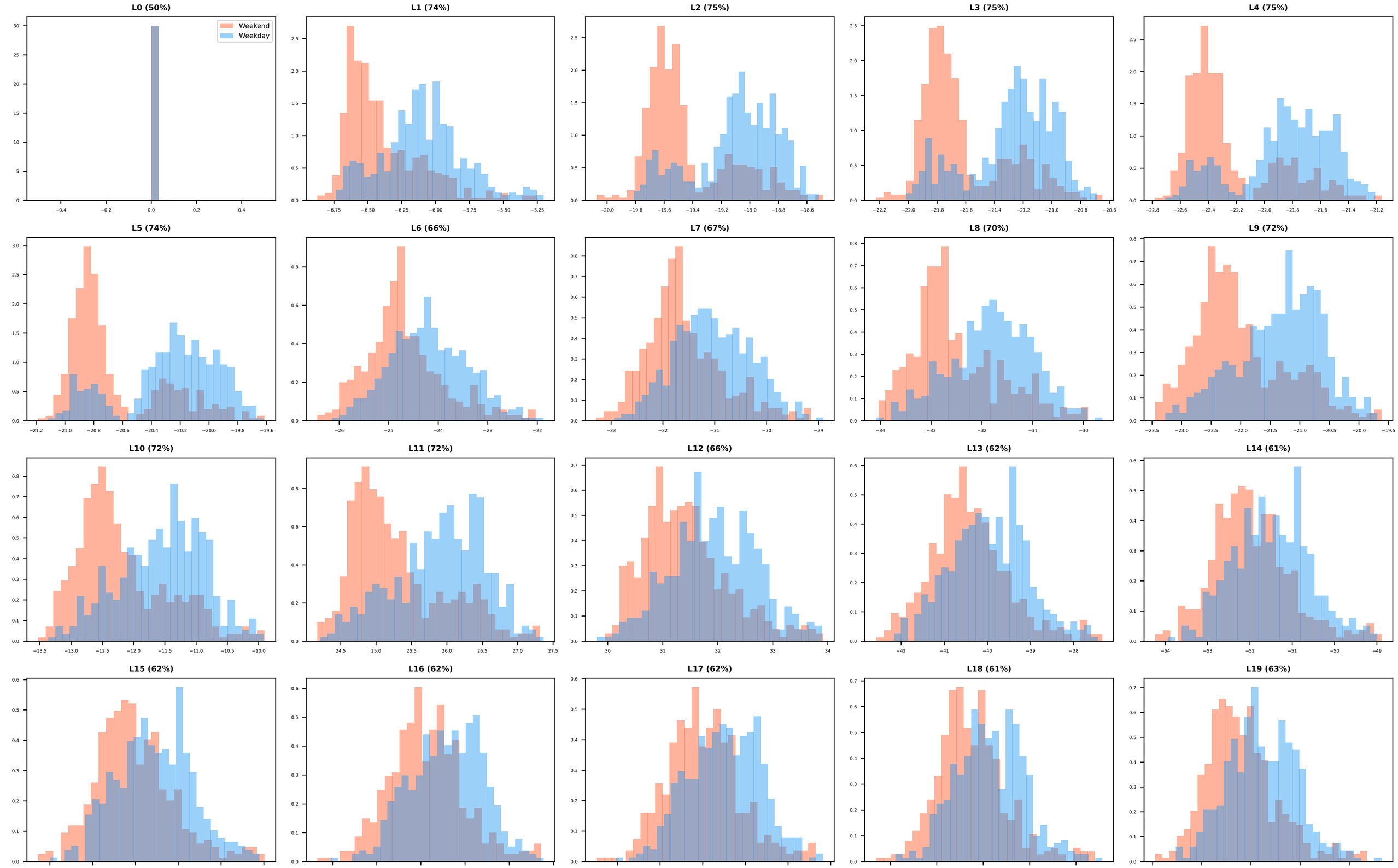
Projection Distribution — V8/exp1a/cfg2 (geo=orig_prompt=cot) — Page 3



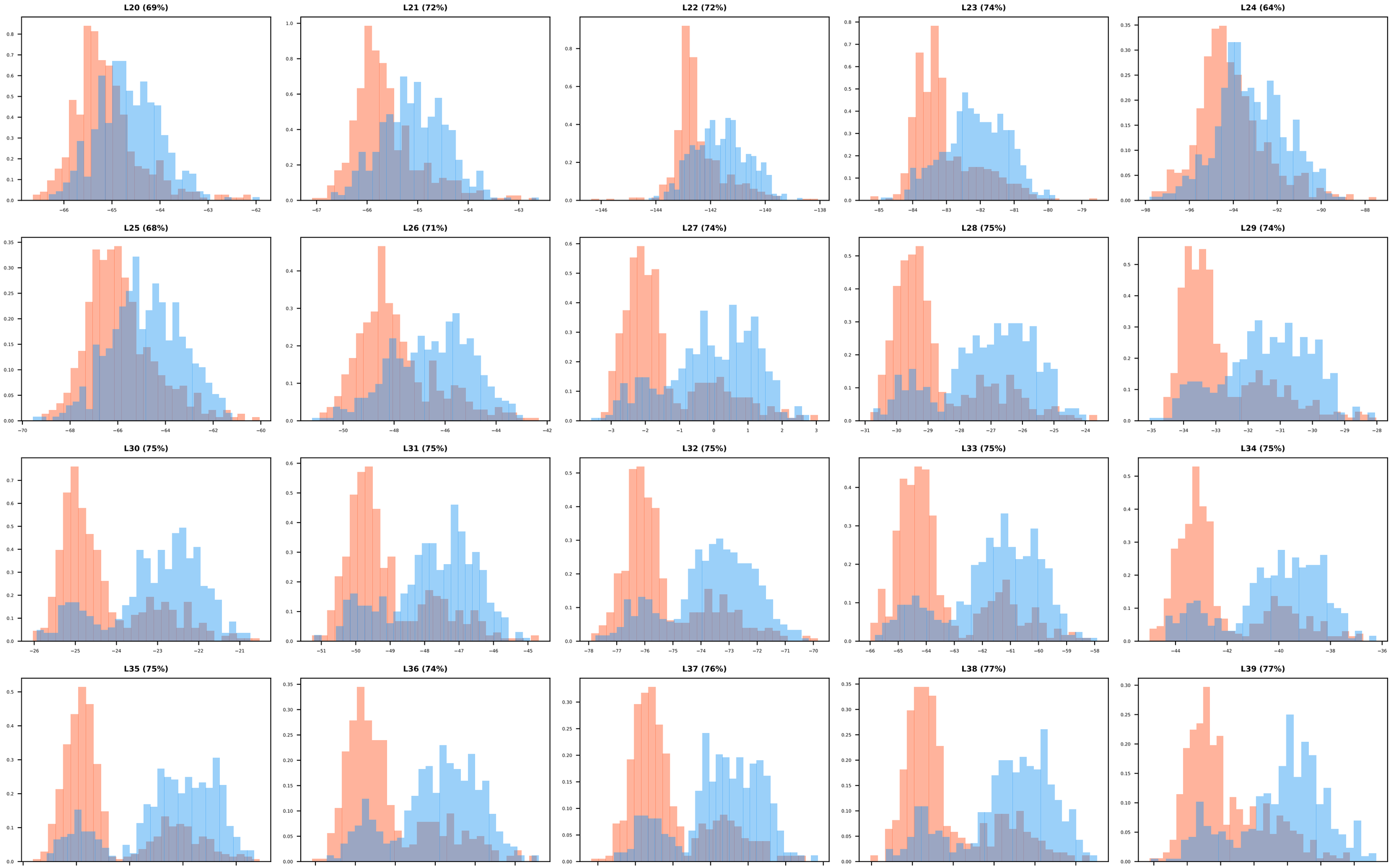
Projection Distribution — V8/exp1a/cfg2 (geo=orig_prompt=cot) — Page 4



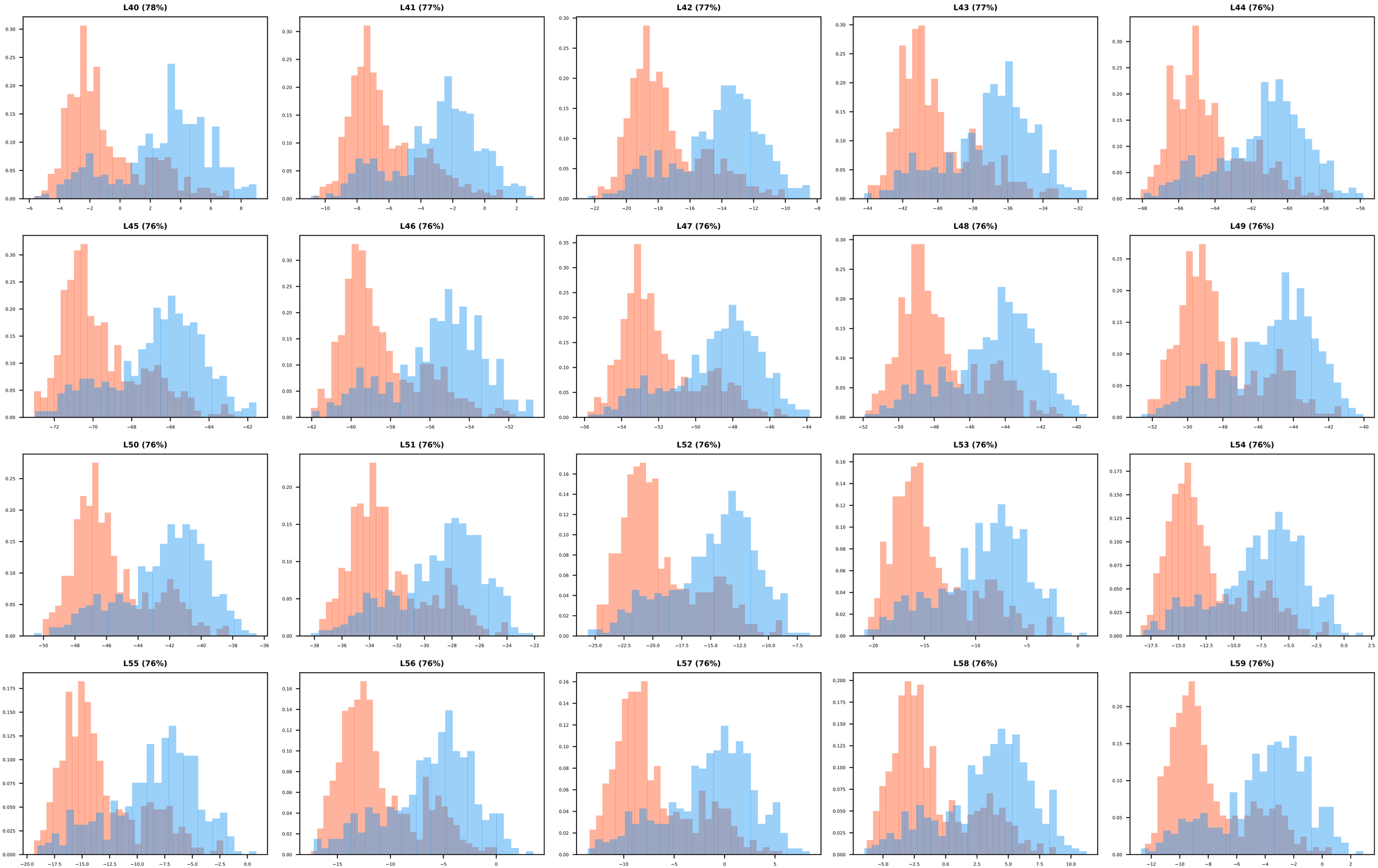
Projection Distribution — V8/exp1a/cfg4 (geo=hash_prompt=cot) — Page 1



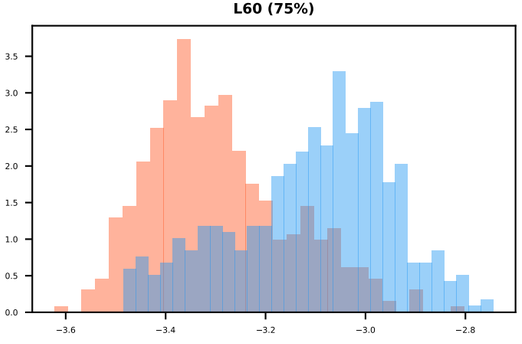
Projection Distribution — V8/exp1a/cfg4 (geo=hash_prompt=cot) — Page 2

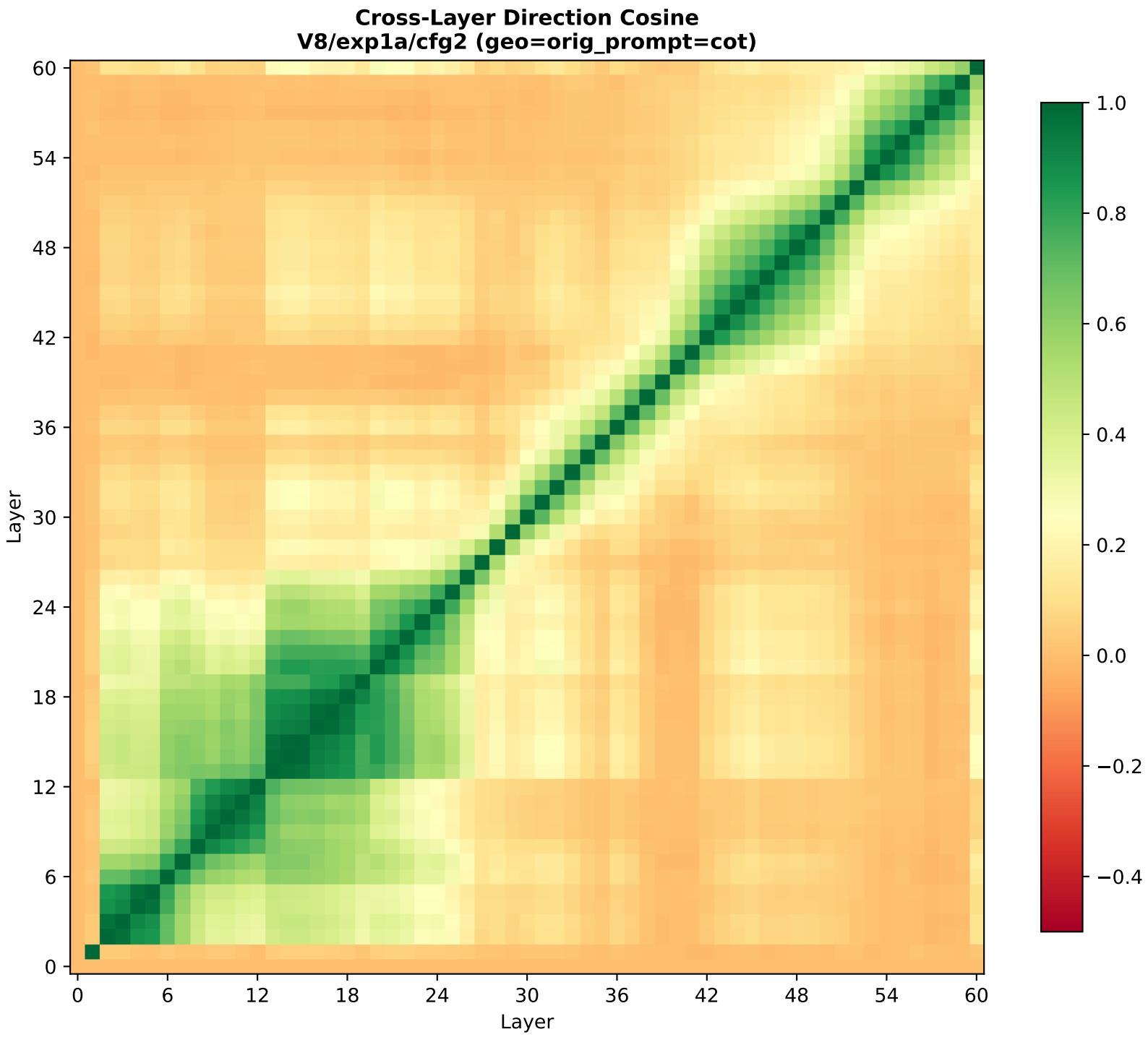


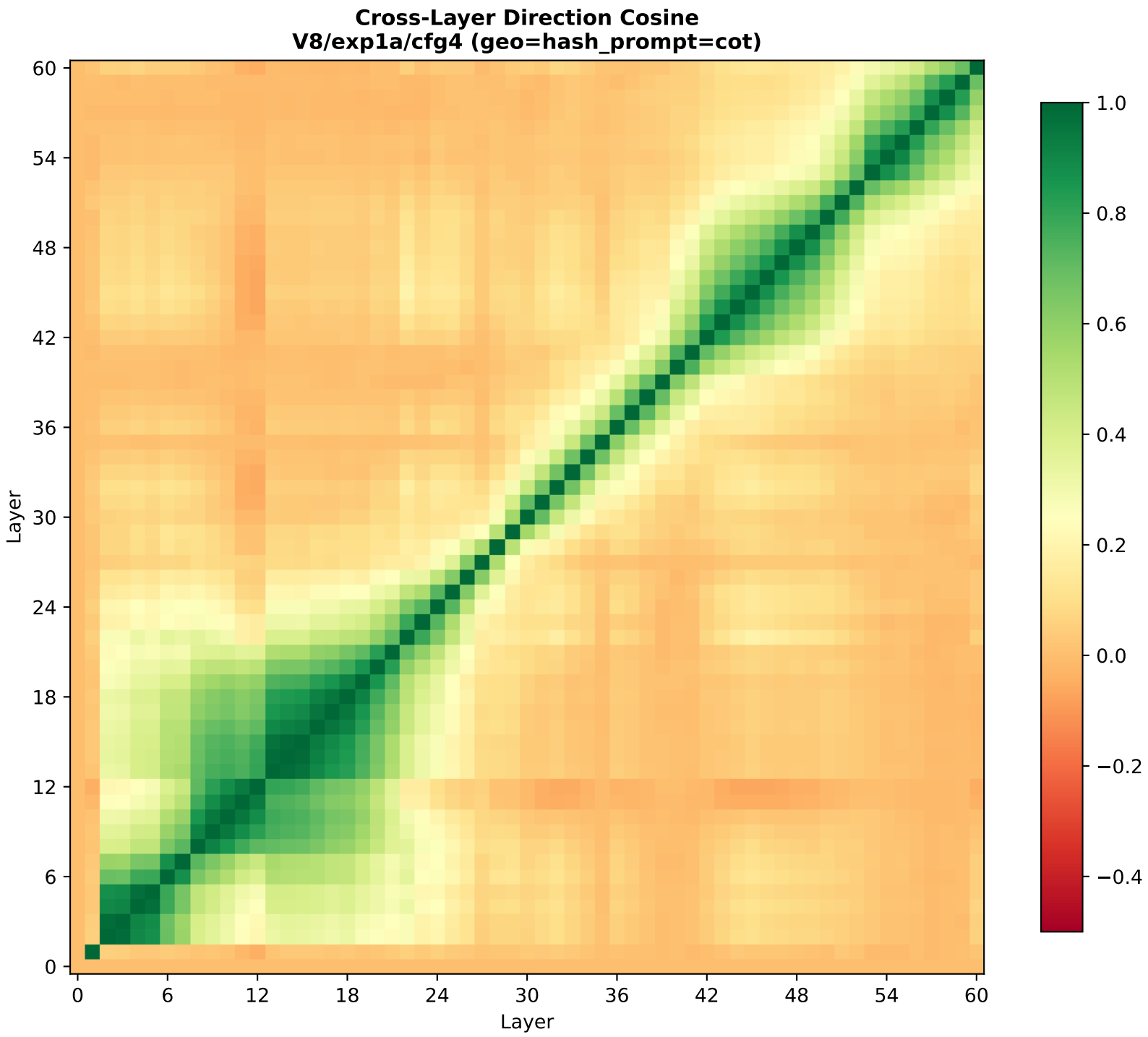
Projection Distribution — V8/exp1a/cfg4 (geo=hash_prompt=cot) — Page 3



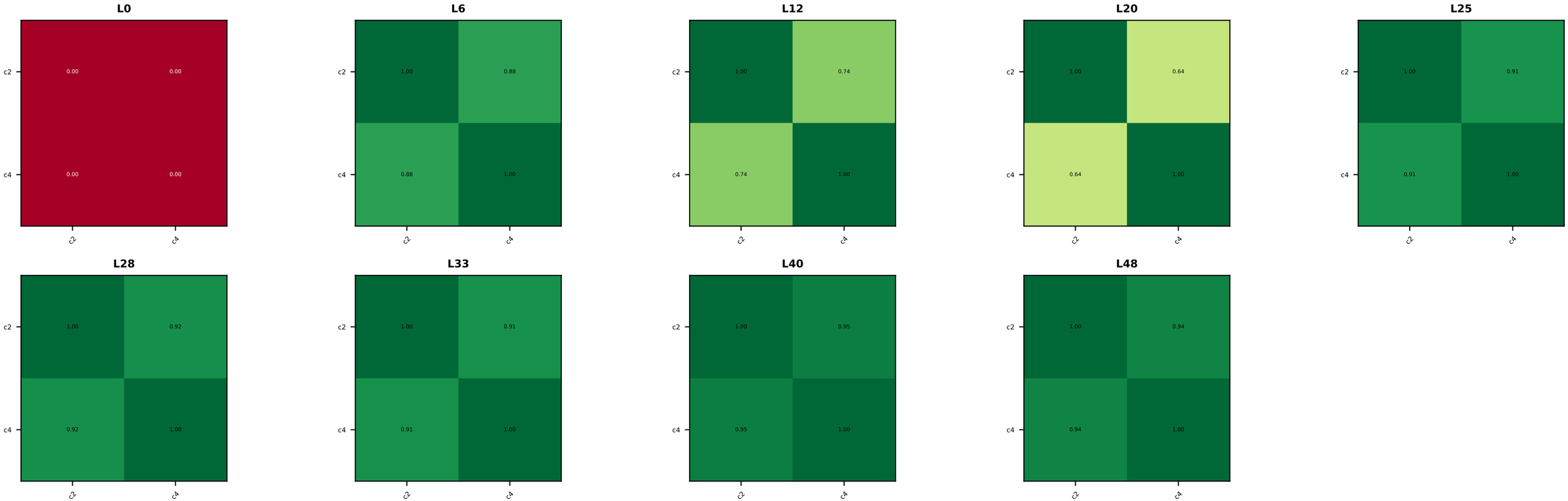
Projection Distribution — V8/exp1a/cfg4 (geo=hash_prompt=cot) — Page 4

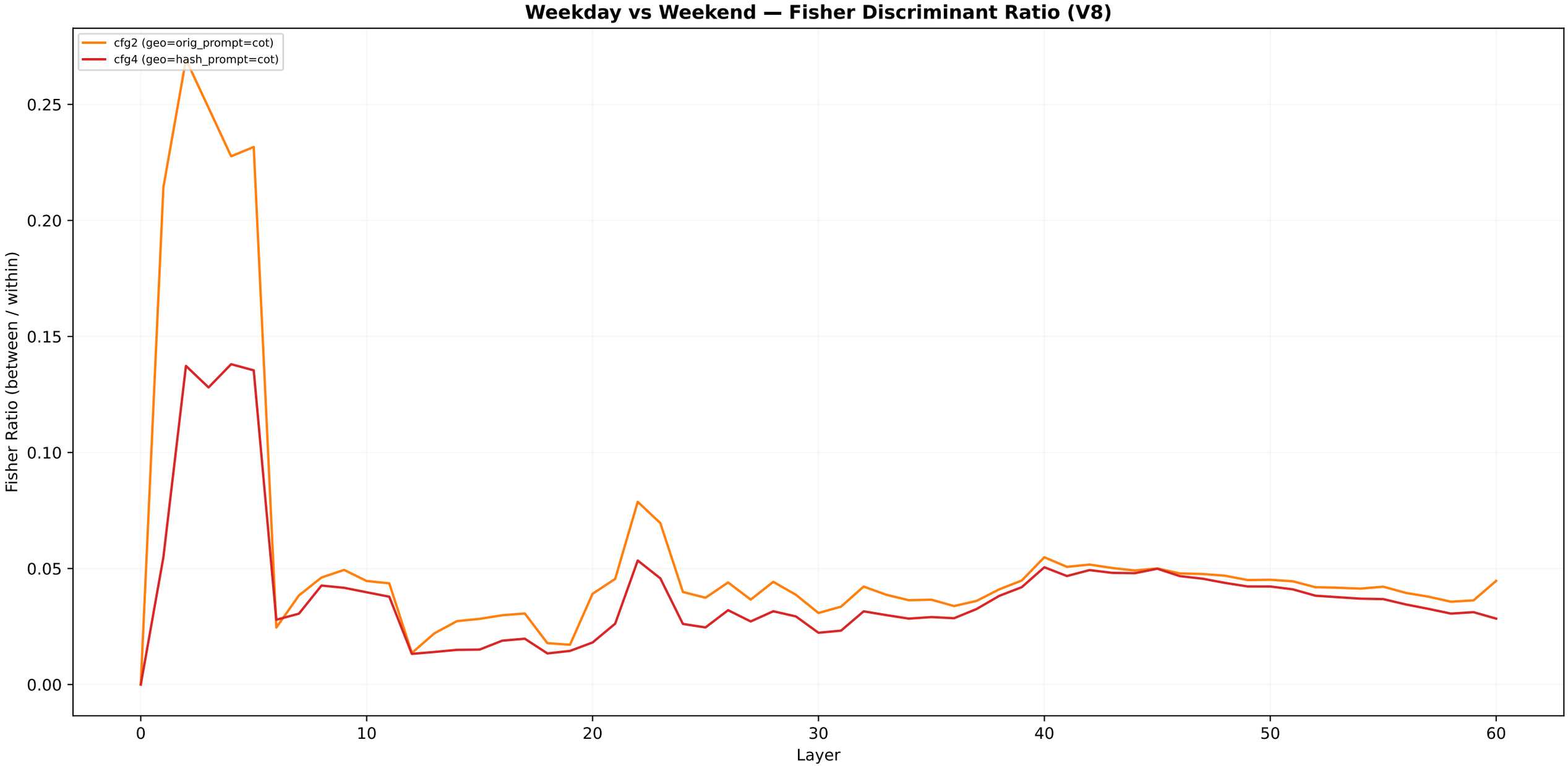


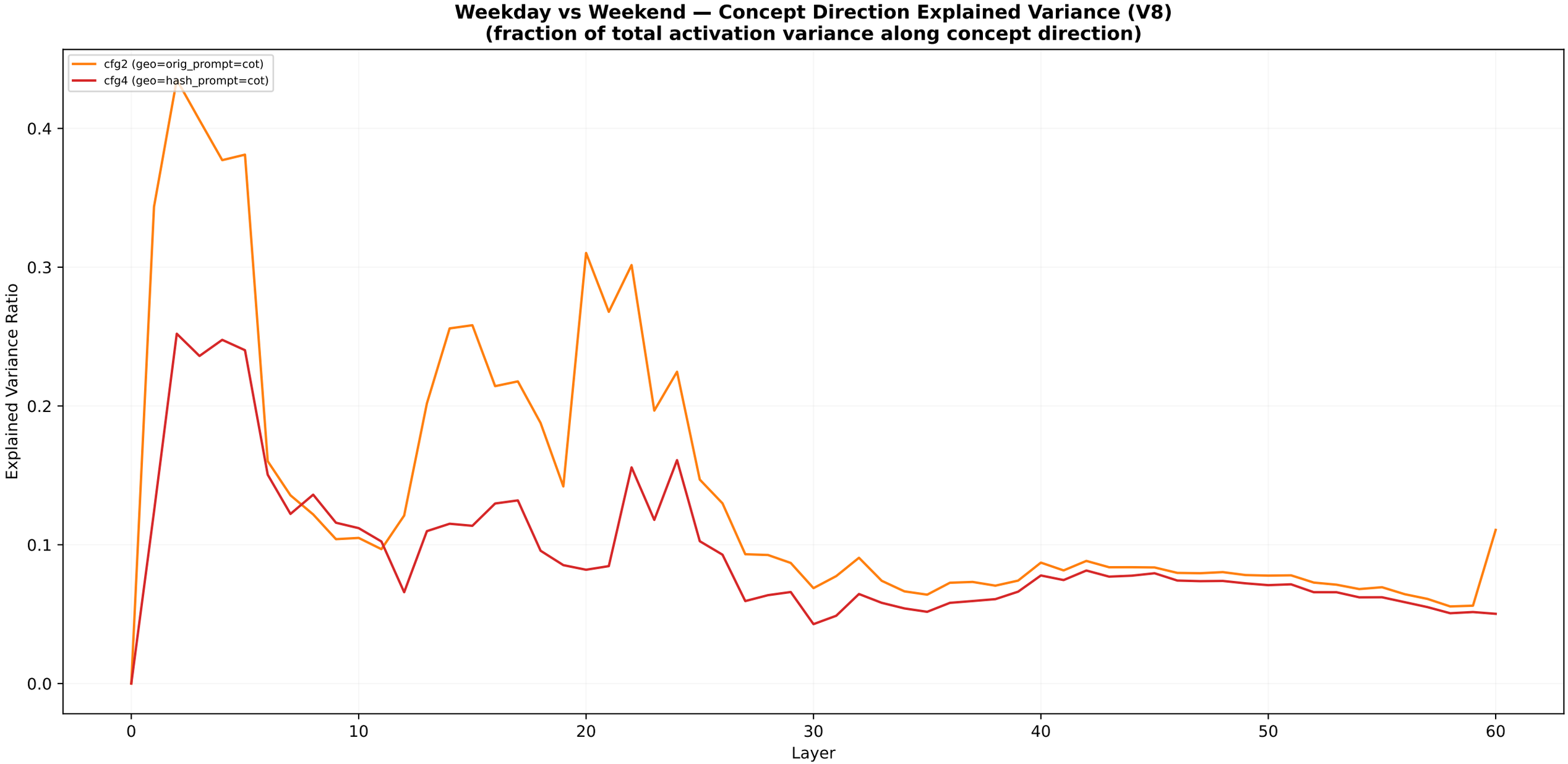

Cross-Layer Direction Cosine
V8/exp1a/cfg2 (geo=orig_prompt=cot)
Layer
Layer
0
6
12
18
24
30
36
42
48
54
60
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

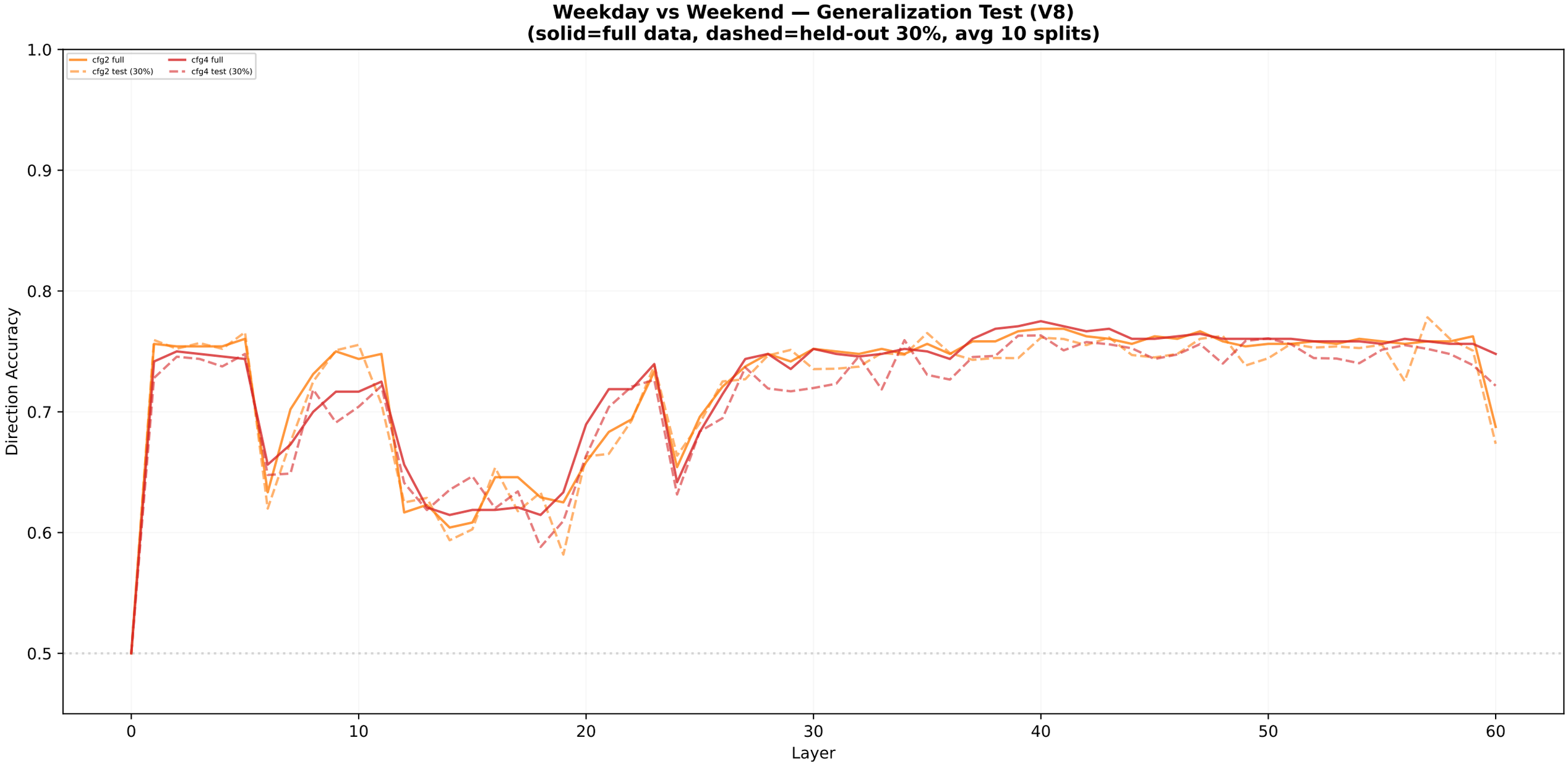

Cross-Layer Direction Cosine
V8/exp1a/cfg4 (geo=hash_prompt=cot)
Layer
Layer
0
6
12
18
24
30
36
42
48
54
60
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

Cross-Config Direction Cosine — V8/exp1a
(do different configs find the same concept direction?)




Weekday vs Weekend — Fisher Discriminant Ratio (V8)
cfg2 (geo=orig_prompt=cot)
cfg4 (geo=hash_prompt=cot)
Fisher Ratio (between / within)
0.25
0.20
0.15
0.10
0.05
0.00
0
10
20
30
40
50
60
Layer


Weekday vs Weekend — Concept Direction Explained Variance (V8)
(fraction of total activation variance along concept direction)
cfg2 (geo=orig_prompt=cot)
cfg4 (geo=hash_prompt=cot)
Explained Variance Ratio
0.0
0.1
0.2
0.3
0.4
0
10
20
30
40
50
60
Layer


Weekday vs Weekend — Generalization Test (V8)
(solid=full data, dashed=held-out 30%, avg 10 splits)
cfg2 full
cfg2 test (30%)
cfg4 full
cfg4 test (30%)
Direction Accuracy
1.0
0.9
0.8
0.7
0.6
0.5
0
10
20
30
40
50
60
Layer

Geometry Summary — V8/exp1a

```
Weekday vs Weekend — Geometry Summary (V8)
==========================================================================================================

 Cfg                  Label     DA  @L      Sep  @L   Fisher  @L      EVR  @L  DA_gen  @L
----------------------------------------------------------------------------------------------------------
   2    geo=orig_prompt=cot  76.9%  40  0.07704  57  0.26936   2 0.435112   2   77.9%  40
   4    geo=hash_prompt=cot  77.5%  40  0.07336  57  0.13804   4 0.252061   2   77.1%  48

----------------------------------------------------------------------------------------------------------
DA: best direction accuracy (all data). Sep: max normalized centroid separation.
Fisher: max between/within variance ratio. EVR: max explained variance ratio.
DA_gen: best generalization DA (70/30 split, avg 5 trials).
```